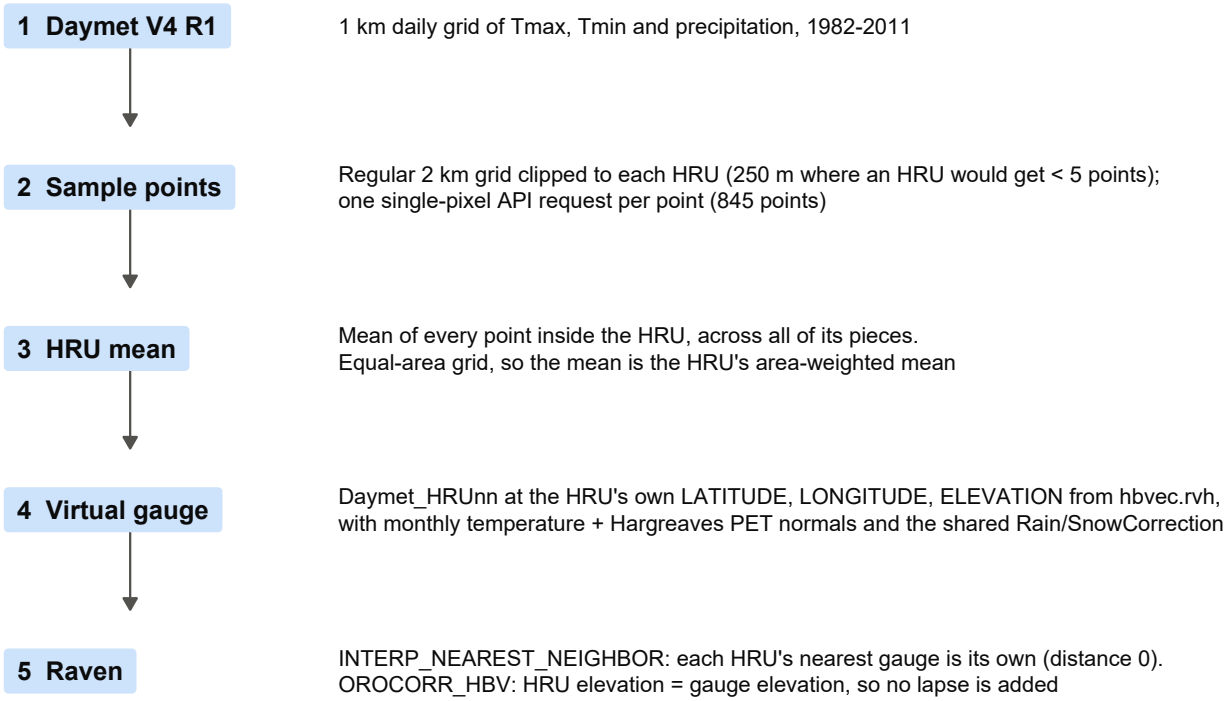
How the Daymet virtual gauges are built

Guichon Creek Raven HBV-EC model · hbvec.rvt · data/scripts/get_daymet.R, write_daymet_rvt.R

The model's precipitation and temperature come from Daymet V4 R1, a 1 km daily gridded dataset. Raven can read gridded NetCDF directly (:GriddedForcing with :GridWeights), but Daymet's NetCDF server now requires a NASA Earthdata login. Instead, the grid is sampled with Daymet's open single-pixel service and turned into one 'virtual gauge' per HRU. The result is the same area-weighted HRU forcing that :GridWeights would give, and the standard Raven build can use it with the .rvi, .rvh and .rvp unchanged.

Raven treats an HRU as one lumped unit with no shape. An HRU that is split into many pieces (as elevation bands usually are) behaves exactly like a single block of the same area. The only spatial question is what forcing each HRU receives, and the answer here is the mean over every piece of the HRU.

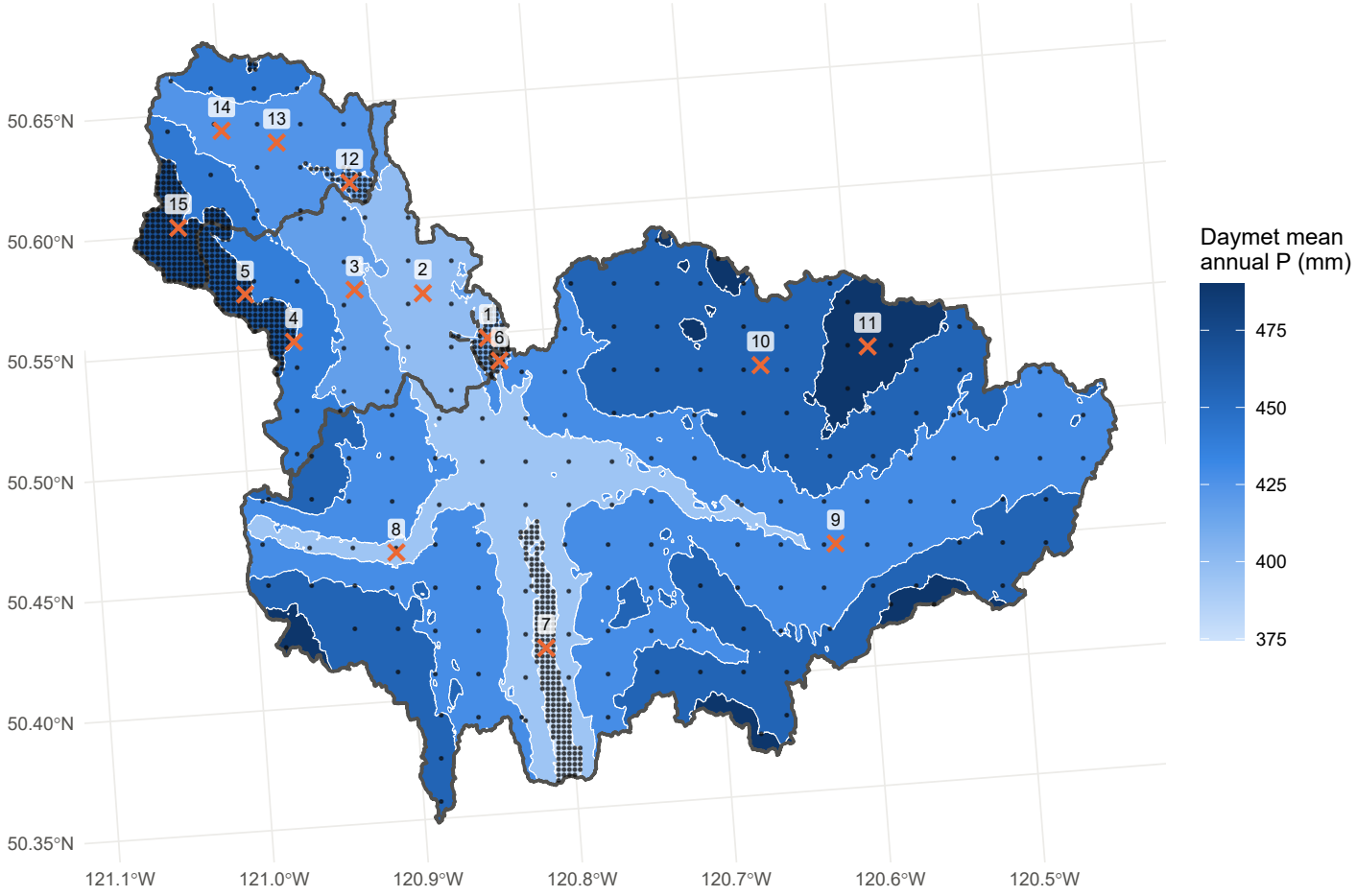
The five steps



Daymet uses a 365-day calendar, omitting Dec 31 in leap years; that day is filled with Dec 30's temperatures and zero precipitation.

HRUs, Daymet sample points and virtual-gauge locations

Fill = HRU-mean annual precipitation. Dots = sample points. × = gauge position (HRU lat/lon in hbvec.rvh). Dark outlines = subbasins 1-3.

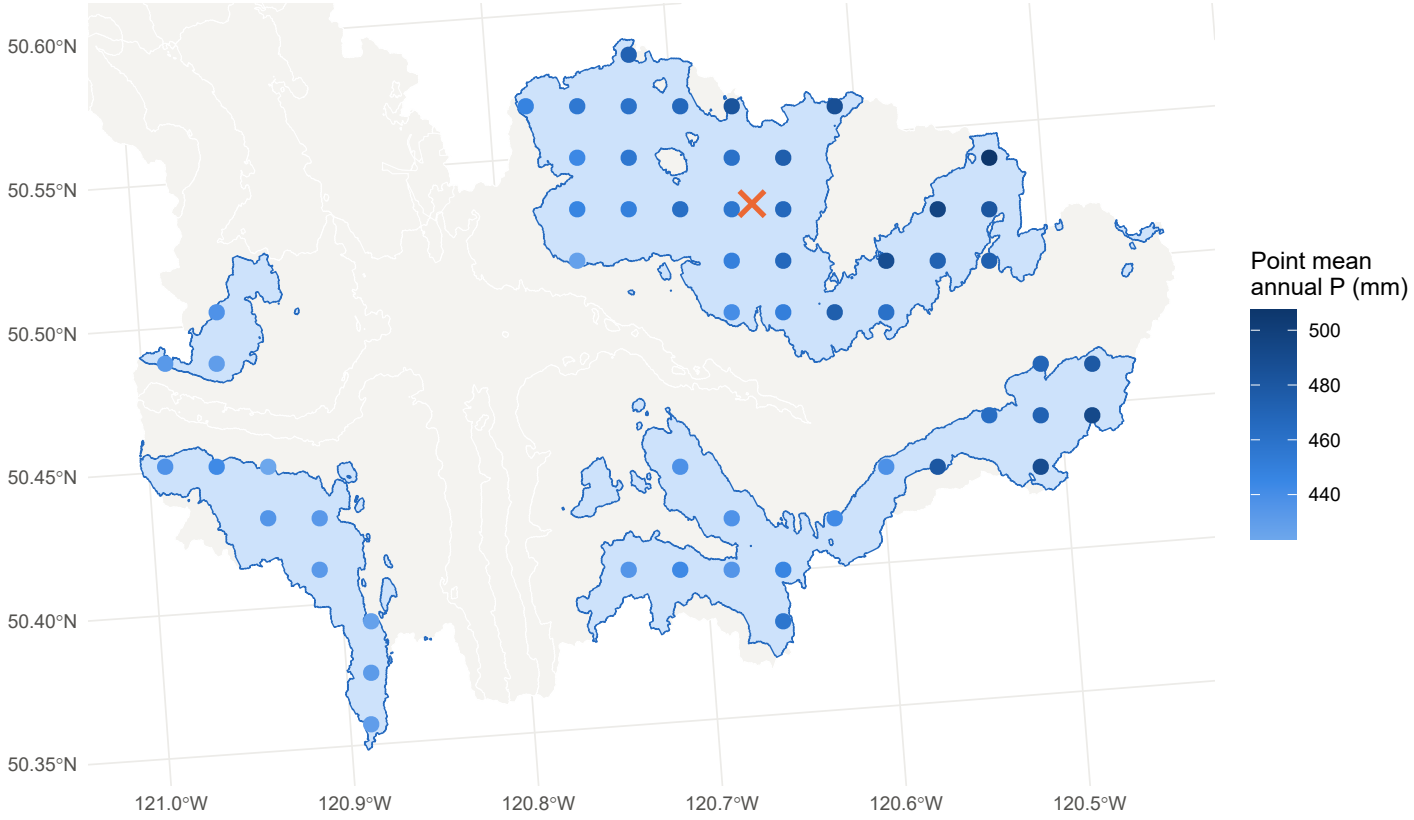


Each elevation-band HRU is scattered across the watershed, so its sample points come from many separate pieces. The densely dotted HRUs (1, 5, 7, 12, 15) are small enough that the 2 km grid would give them fewer than 5 points, so they are sampled every 250 m instead. That is finer than Daymet's 1 km pixels, so neighbouring points repeat values, which does not bias the mean.

The × marks are the gauges' lat/lon from hbvec.rvh, which is each HRU's representative location. Because every HRU has its own gauge at exactly its own coordinates, Raven's nearest-neighbour step always picks that gauge. Precipitation rises toward the east and at higher elevation; Daymet captures both, whereas the single Lornex station had to be lapsed by elevation alone.

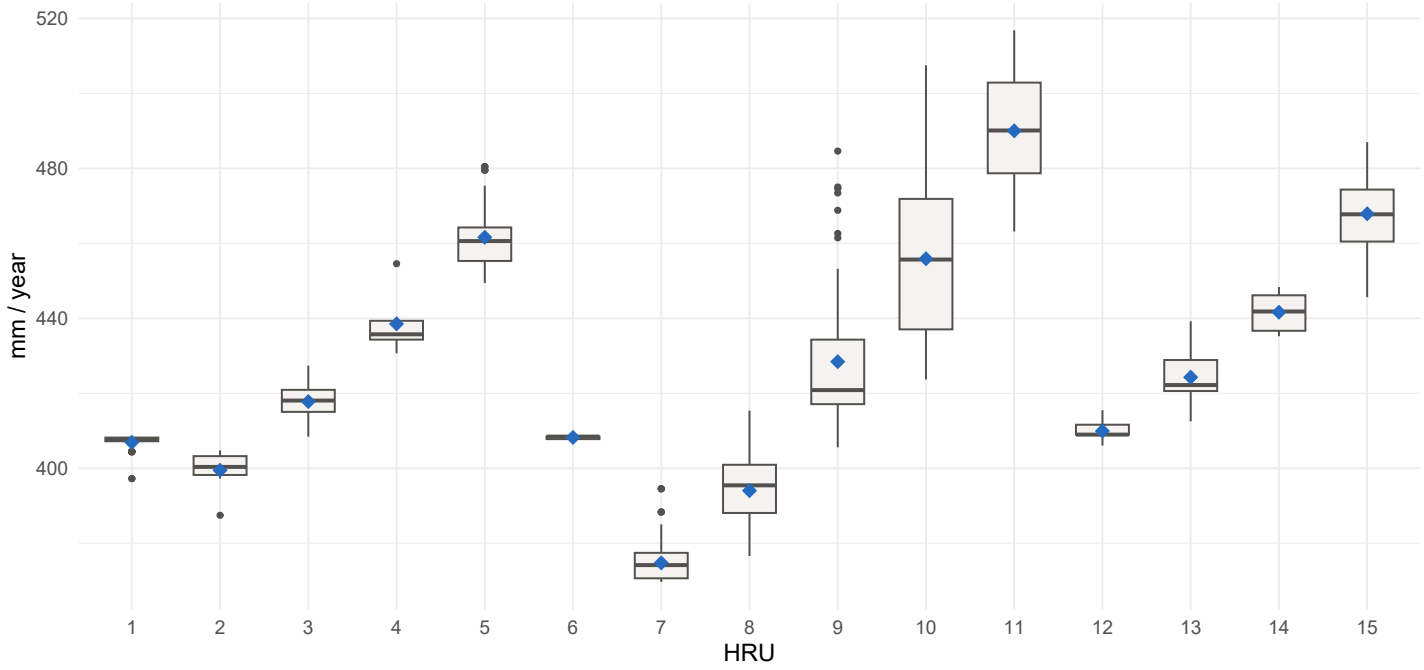
Close-up: HRU 10 (subbasin 2, 1400-1600 m band), 54 pieces, 57 sample points

All pieces feed one HRU mean (456 mm/yr). The gauge × only labels the HRU for Raven's nearest-neighbour match; its position does not affect the forcing values.



Spread of mean annual precipitation among the sample points in each HRU

Box = points in the HRU. Blue diamond = HRU mean written to the virtual gauge.



Per-HRU summary

Points = sample points. Missed % = share of HRU area in pieces too small to contain a point. P and T are 1982-2011 means over the points.

HRU	SB	Area km²	Elev m	Pieces	Points	Missed %	P mm/yr	P range	P CV %	T range °C
1	1	3.2	1215	1	50	0.0	407	397-409	0.6	0.5
2	1	37.2	1151	5	9	0.0	399	387-405	1.3	0.8
3	1	32.0	1292	70	8	11.6	418	408-427	1.5	0.9
4	1	25.6	1478	51	6	0.1	439	431-455	1.9	1.2
5	1	10.9	1692	102	175	0.2	462	449-481	1.8	1.6
6	2	0.8	1232	1	12	0.0	408	408-409	0.1	0.0
7	2	11.8	974	1	186	0.0	375	370-395	1.3	1.2
8	2	83.2	1122	14	24	0.3	394	377-415	2.4	1.5
9	2	250.1	1309	29	62	0.3	428	406-485	4.4	1.4
10	2	235.6	1484	54	57	2.2	456	424-508	4.5	1.6
11	2	38.7	1670	20	9	5.7	490	463-517	3.6	0.7
12	3	2.1	1186	4	34	0.3	410	406-415	0.6	0.4
13	3	39.1	1308	2	10	0.1	424	413-439	2.0	1.2
14	3	22.5	1494	8	6	0.3	442	435-448	1.3	0.7
15	3	12.2	1743	4	197	0.0	468	446-487	2.0	2.2

What this means

Splitting an HRU into pieces does not bias its forcing: every piece contributes points, and the HRU's gauge carries their mean. The gauge's map position is only a label for Raven's nearest-neighbour match. In one case (HRU 14) it falls outside the HRU's own polygon, which has no effect.

Some slivers are smaller than the 2 km grid spacing and get no point (largest gap: HRU 3, 12% of its area). They lie in the same elevation band as the rest of the HRU, where precipitation varies by about 1-2%, so the error is about 0.2%. Sampling Daymet's own 1 km pixels with exact overlap weights would remove it.

The real limit is lumping itself, not fragmentation. Within an HRU, mean annual precipitation varies by up to 4-5% (CV), about ±10% end to end (HRUs 9 and 10, likely west to east), and mean temperature by up to about 2 °C. One HRU mean smooths this, which slightly affects rain/snow partitioning and melt on days near 0 °C. The 200 m elevation bands keep the temperature range small. Daymet is itself interpolated from stations, so the true spread is probably larger than shown.

The solar-site HRUs (1 and 6) are single pieces with nearly uniform forcing (P CV 0.6% and 0.1%), so the baseline-versus-solar difference is unaffected.